Gantry Controller Board

Three STM32 development-board candidates | Quick comparison + technical notes

Purpose: identify three alternatives for controlling four gantry motor drivers, reading sensors, and communicating with a second controller board. This document is a research shortlist, not a purchase authorization or completed wiring design.

1. Quick comparison

Board	Best reason to consider	Pros	Cons / concerns
NUCLEO-G474RE	Best overall value	Motor-control-oriented MCU; 6 serial peripherals on chip; low cost; approachable tools	64-pin board; less room for wiring; detailed pin-conflict check needed
NUCLEO-H723ZG	Most expansion room	144-pin board; high performance; more accessible I/O; Ethernet available	Higher cost; higher complexity; more capability than likely necessary
NUCLEO-F446RE	Budget-friendly backup	Mature STM32 platform; affordable; 6 serial peripherals on chip	Older platform; 64-pin constraints; legacy Mini-B connection

Plain-language takeaway

G474RE: The sensible starting choice if it can fit every UART connection and sensor on separate, accessible pins.

H723ZG: The safer choice for expansion if the team expects many more inputs, extra features, or complicated wiring.

F446RE: A capable inexpensive alternative, but with fewer reasons to choose it over the G474RE.

What the team actually needs

- Four external motor drivers, preferably with one dedicated hardware UART per driver.
- One additional controller-to-controller communication link (UART proposed, not yet required).
- GPIO for limit switches, motor enables, homing, fault inputs, and possible load/position sensors.
- Onboard programming/debugging, user button and LED; USB connection; regulated wall power.
- A separately wired emergency stop that can disable motor power without depending on firmware.

2. Detailed option: NUCLEO-G474RE

Microcontroller: STM32G474RE, Arm Cortex-M4 up to 170 MHz, 512 KB flash, 128 KB SRAM. [1, 2]

Communications: The MCU family lists five USART/UART instances plus one low-power UART (six serial instances in total), along with I^2C, SPI, FDCAN, and USB capabilities. [2]

Board features: Nucleo-64 format, Arduino Uno V3 / ST Morpho expansion, onboard ST-LINK-V3E debugging, user LED and button. ST-LINK uses a USB Micro-B connector. [1, 3]

Why it fits: STM32G4 is especially relevant to motor-control work. It offers timing and control peripherals while keeping the cost and software setup manageable for a student team. [2]

Limitations: A 64-pin chip means multiple peripherals compete for limited pins. The ST-LINK virtual COM-port routing and solder-bridge configuration also need review before reserving five UARTs. [3]

Our assessment: Recommended as the first cost-conscious candidate, conditional on an accepted pin-allocation plan.

3. Detailed option: NUCLEO-F446RE

Microcontroller: STM32F446RE, Arm Cortex-M4 up to 180 MHz, 512 KB flash, about 128 KB general SRAM. [4, 5]

Communications: The F446 family includes four USARTs and two UARTs (six serial instances), three I^2C controllers, CAN, and multiple timers. [5]

Board features: Nucleo-64 format, Arduino Uno V3 / ST Morpho expansion, onboard ST-LINK, user button and LED. Its ST-LINK connector is USB Mini-B. [4, 6]

Why it fits: Mature documentation, straightforward STM32CubeIDE/HAL support, and enough processor capability for coordinating a gantry controller if pin mapping works. [4–6]

Limitations: As with the G474RE, six MCU serial interfaces do not necessarily equal six independently usable board-level ports. Fewer expansion pins than a 144-pin Nucleo, and older board connectivity. [5, 6]

Our assessment: Worth keeping as an affordable alternative, but not our top pick without a specific cost or familiarity advantage.

4. Detailed option: NUCLEO-H723ZG

Microcontroller: STM32H723ZG, Arm Cortex-M7 up to 550 MHz, 1 MB flash, 564 KB total SRAM. [7, 8]

Communications: ST lists up to six USART/UART/LPUART interfaces. The device also includes I^2C, SPI, FDCAN, Ethernet capability, and many timer resources. [8]

Board features: Nucleo-144 format with Zio / Morpho expansion headers, three user LEDs, user/reset buttons, and onboard STLINK-V3E. Debug connector is USB Micro-B; the board also has a separate USB Micro-AB user connection. [7, 9]

Why it fits: More physical pins and expansion options can reduce pin-routing pressure for four motor UARTs, a fifth serial connection, and multiple sensors. It also leaves room for future features. [7–9]

Limitations: The additional CPU performance is not itself needed to send UART commands. Firmware setup and system integration may be more involved; it costs more than the Nucleo-64 options. [7–9]

Our assessment: Best expansion-oriented candidate. Choose it if the G474RE pin map is too crowded or future I/O growth is a priority.

5. Side-by-side technical snapshot

Property	G474RE	H723ZG	F446RE
MCU core / max clock	Cortex-M4 / 170 MHz	Cortex-M7 / 550 MHz	Cortex-M4 / 180 MHz
Flash / RAM	512 KB / 128 KB	1 MB / 564 KB	512 KB / ~128 KB
Chip serial instances	6 total	Up to 6 total	6 total
Board size	Nucleo-64	Nucleo-144	Nucleo-64
ST-LINK USB	Micro-B	Micro-B	Mini-B
Cost position	Budget	Higher	Budget
4 motor UARTs + extra link	Possible; unverified pins	Possible; unverified pins	Possible; unverified pins

All three support the STM32Cube development ecosystem, but none should be declared wiring-ready until the UART/USB/debug assignments and sensor GPIOs are checked on the exact board revision. [1–9]

6. Before we choose a board

To convert this shortlist into a defensible final selection, the team should complete the following checks:

- Confirm each motor-driver model and whether it truly requires full-duplex UART, half-duplex UART, STEP/DIR pulses, an enable pin, or other signals.
- Count all sensors and define their interfaces: plain digital inputs, analog inputs, encoder channels, I²C, SPI, or other.
- Decide the board-to-board protocol: UART, CAN, RS-485 transceiver, or another interface; identify cable length and noise environment.
- Create one STM32CubeMX project per candidate; reserve four independent motor UARTs plus a fifth board-link interface, then check ST-LINK VCP and all sensor pins.
- Cross-check the CubeMX pin selection against the exact board manual, schematic, accessible connector pins, and any solder-bridge restrictions.
- Choose the correct regulated board supply and motor-driver power supply; connect grounds and signal-level conversion as required.
- Implement a hardware emergency stop / driver-disable path independent of the MCU; firmware safety checks are additional safeguards.
- Check live vendor pricing and inventory before ordering; budget for connectors, motor drivers, level shifters, and wiring.

7. Preliminary recommendation

Start with the NUCLEO-G474RE for its value and motor-control feature set. Keep the NUCLEO-H723ZG as the stronger expansion option if the pin map becomes crowded. Retain the NUCLEO-F446RE as a budget backup. This ranking is provisional: compatibility with the final driver and sensor list outweighs CPU speed or headline peripheral counts.

References — STMicroelectronics

[1] NUCLEO-G474RE board — https://www.st.com/en/evaluation-tools/nucleo-g474re.html

[2] STM32G474RE MCU specifications — https://www.st.com/en/microcontrollers-microprocessors/stm32g474re.html

[3] UM2505: G4 Nucleo-64 board user manual — https://www.st.com/resource/en/user_manual/um2505-getting-started-with-stm32g4-series-nucleo64-board-stmicroelectronics.pdf

[4] NUCLEO-F446RE board — https://www.st.com/en/evaluation-tools/nucleo-f446re.html

[5] STM32F446 MCU family — https://www.st.com/en/microcontrollers-microprocessors/stm32f446.html

[6] UM1724: Nucleo-64 board user manual — https://www.st.com/resource/en/user_manual/um1724-stlinkv21-in-circuit-debuggerprogrammer-for-stm8-and-stm32-stmicroelectronics.pdf

[7] NUCLEO-H723ZG board — https://www.st.com/en/evaluation-tools/nucleo-h723zg.html

[8] STM32H723ZG MCU specifications — https://www.st.com/en/microcontrollers-microprocessors/stm32h723zg.html

[9] UM2407: H7 Nucleo-144 board user manual — https://www.st.com/resource/en/user_manual/um2407-stm32h7-nucleo144-board-stmicroelectronics.pdf